

Reply to Nick — Čech complex definition

Draft. Sending after Lean revert + paper §2.2 rewrite are in flight so the commitment is backed by code, not just a promise.

Hi Nick,

You're right, and the remark doesn't address it. The definition as currently written isn't a simplicial complex — under the nerve-only convention I had (triple intersection of $B(x_i, r)$), $F_{\{ijk\}} = 1$ only forces pairwise distance $\leq 2r$, while edges require $\leq r$, so downward closure fails.

The issue is that this convention drifted away from both my thesis and Kahle's standard reference (K10, Def 1.4): the random Čech complex is the nerve of $\{B(x_i, r/2)\}_i$, not $\{B(x_i, r)\}$. With $r/2$ -balls and edges at $\leq r$, triple intersection forces pairwise distance $\leq r$, so every filled triangle has its three boundary edges — genuine simplicial complex, downward closed, matches the literature.

I'm reverting to the Kahle/thesis convention:

- $\sigma \in \text{Cech}(n, r, d)$ iff $\bigcap_{i \in \sigma} B(x_i, r/2) \neq \emptyset$
- edges at $d_{\infty}(x_i, x_j) \leq r$ (unchanged — matches $p = (2r)^d$ on the torus)

This invalidates a chunk of the moment machinery (the leading-order constant in `fillingProb` was $(12r^2)^d$ under the wrong convention; it becomes $\sim (3r^2/4)^d$ under the correct one, and the $(7r/2)^d$ wedge term in the deep covariance formula shifts correspondingly). The Lean catalog needs re-derivation — `centered_edge_moment_fill`, `geometricCov_eq_deep`, the TorusIntegrals volume lemmas, and the sparse-regime + τ_{ff} analysis in §5. Detection threshold ($n^{3/2} \cdot \text{geomCov} \rightarrow \infty$) and the overall TV $\rightarrow 1$ result are unaffected in form, just the explicit constants change.

I'll resubmit the revised draft with corrected Def 2.3, new Remark 2.4 citing Kahle Def 1.4, and the updated constants once the Lean refactor lands. Estimating about a week (~5–10 Aristotle jobs).

Apologies for the noise — this is a real bug, not a definitional preference.

David